

Multiplication of polynomials

Every term of the first times every term of the second — and a rectangle explains why.

LESSON 73–74

2 × 40 MIN

MATEMATIKA 7, §34

S8 2.3

LESSON CLOCK

15'

25'

25'

20'

5'

By a monomial	15 min
Two binomials	25 min
The area picture	25 min
Longer products	20 min
Homework	5 min

LEARNING OBJECTIVES

- Multiply a polynomial by a monomial.
- Multiply two binomials and collect the result.
- Multiply a binomial by a trinomial.
- Use an area diagram to justify the expansion.

TEXTBOOK

National	pp. 205–211
Cambridge	Stage 8, pp. 18–24
Workbook	Exercise 2.3

01

Explanation

By a monomial

$$k(a + b + c) = ka + kb + kc$$

Multiply every term — including the last, and including the signs.

PRODUCT	EXPANDED
$3x(x - 4)$	$3x^2 - 12x$
$-2a(a^2 - 3a + 1)$	$-2a^3 + 6a^2 - 2a$
$xy(x + y)$	$x^2y + xy^2$

THE LAST TERM IS THE ONE MOST OFTEN FORGOTTEN

In $-2a(a^2 - 3a + 1)$ the third term gives $-2a$. Counting the terms before and after expanding catches the omission.

Two binomials

$$(a + b)(c + d) = ac + ad + bc + bd$$

Four products, one for each pair. Then collect any like terms.

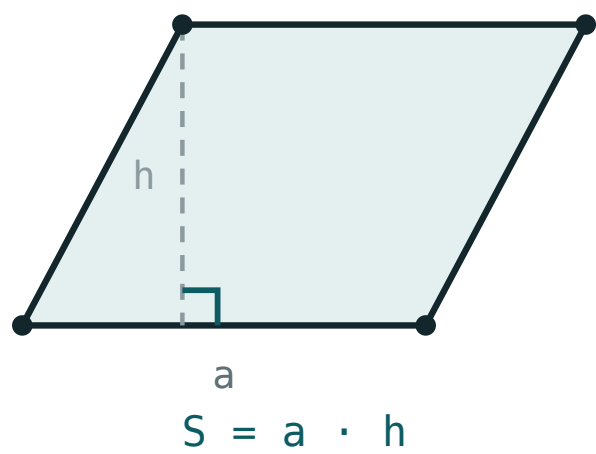
PRODUCT	FOUR TERMS	COLLECTED
$(x + 2)(x + 3)$	$x^2 + 3x + 2x + 6$	$x^2 + 5x + 6$
$(x - 2)(x + 5)$	$x^2 + 5x - 2x - 10$	$x^2 + 3x - 10$
$(2a - 1)(a + 4)$	$2a^2 + 8a - a - 4$	$2a^2 + 7a - 4$
$(x - 3)(x + 3)$	$x^2 + 3x - 3x - 9$	$x^2 - 9$

THE LAST ROW IS WORTH REMEMBERING

The middle terms cancel and only $x^2 - 9$ survives. That pattern — the difference of two squares — is the subject of a whole lesson in Quarter III.

The area picture

A rectangle of sides $a + b$ and $c + d$ splits into four smaller rectangles.



A rectangle cut into four — the four products of the expansion.

	C	D
a	ac	ad
b	bc	bd

The whole area is the sum of the four parts — which is exactly the expansion. The grid also makes it impossible to miss a product.

USE THE GRID WHEN THERE ARE MORE THAN FOUR PRODUCTS

For a binomial times a trinomial there are six, and a 2×3 grid keeps them all in view.

Longer products

Example. $(x + 2)(x^2 - 3x + 1)$. Six products:

	X^2	$-3X$	$+1$
x	x^3	$-3x^2$	x
$+2$	$2x^2$	$-6x$	2

$$x^3 - 3x^2 + 2x^2 + x - 6x + 2 = x^3 - x^2 - 5x + 2$$

THE DEGREE OF A PRODUCT IS THE SUM OF THE DEGREES

Here $1 + 2 = 3$, and the answer is cubic. Checking the degree first tells you what to expect before any arithmetic.

02 Worked examples

EXAMPLE 1 Expand $-2a(a^2 - 3a + 1)$.

① $-2a \times a^2 = -2a^3$

② $-2a \times (-3a) = 6a^2$
Two minuses.

③ $-2a \times 1 = -2a$

④ $= -2a^3 + 6a^2 - 2a$
Three terms in, three out.

ANSWER

$$-2a^3 + 6a^2 - 2a$$

EXAMPLE 2 Expand $(x - 2)(x + 5)$.

① $x \cdot x = x^2, x \cdot 5 = 5x.$

② $-2 \cdot x = -2x, -2 \cdot 5 = -10.$

③ $x^2 + 5x - 2x - 10$

④ $= x^2 + 3x - 10$

ANSWER

$$x^2 + 3x - 10$$

EXAMPLE 3 Expand $(x + 2)(x^2 - 3x + 1)$.

① x row: $x^3 - 3x^2 + x$.

② $+2$ row: $2x^2 - 6x + 2$.

③ Add and collect.

④ $= x^3 - x^2 - 5x + 2$
Degree 3 ✓

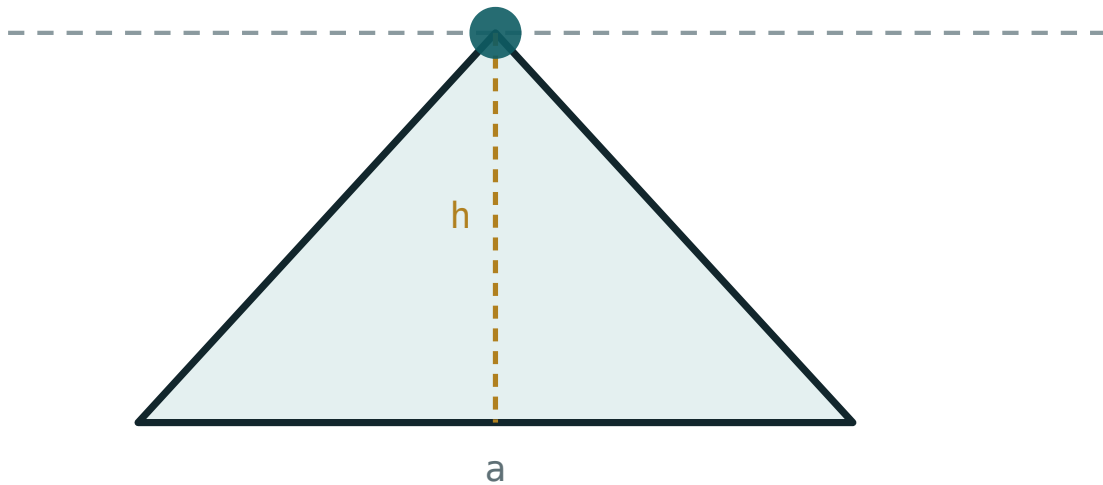
ANSWER

$$x^3 - x^2 - 5x + 2$$

03 Interactive model

Draw the rectangle for $(x + 2)(x + 3)$ with its four parts labelled; the expansion is read off the picture before any rule is stated.

Expanding as an area

base a **220**height h **120**area $\frac{1}{2}ah$ **13200**perimeter **545.6**
04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
To multiply out	Ko'paytirish	Умножение
To expand	Yoyish	Раскрыть
Binomial	Ikki had	Двучлен
Trinomial	Uch had	Трёхчлен
Every term	Har bir had	Каждый член
Area diagram	Yuza chizmasi	Схема площадей
To collect	Ixchamlash	Приведение подобных
Product	Ko'paytma	Произведение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$3x(x - 4)$ equals:

$(a + b)(c + d)$ gives how many products?

$(x + 2)(x + 3)$ equals:

$(x - 3)(x + 3)$ equals:

A binomial times a trinomial gives:

The degree of a product is:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $3x(x - 4)$

ANSWER $3x^2 - 12x$

2. $2a(a + 5)$

ANSWER $2a^2 + 10a$

3. $xy(x + y)$

ANSWER $x^2y + xy^2$

4. $(x + 2)(x + 3)$

ANSWER $x^2 + 5x + 6$

5. $(x + 1)(x + 4)$

ANSWER $x^2 + 5x + 4$

6. $(x - 3)(x + 3)$

ANSWER $x^2 - 9$

7. $(x + 5)(x - 5)$

ANSWER $x^2 - 25$

Medium · the standard question

7 PROBLEMS

1. $-2a(a^2 - 3a + 1)$

ANSWER $-2a^3 + 6a^2 - 2a$

2. $(x - 2)(x + 5)$

ANSWER $x^2 + 3x - 10$

3. $(2a - 1)(a + 4)$

ANSWER $2a^2 + 7a - 4$

4. $(3x + 2)(x - 1)$

ANSWER $3x^2 - x - 2$

5. $(x + 2)(x^2 - 3x + 1)$

ANSWER $x^3 - x^2 - 5x + 2$

6. $(a - b)(a + b)$

ANSWER $a^2 - b^2$

7. $(2x - 3)(2x + 3)$

ANSWER $4x^2 - 9$

Hard · stretch and reason

7 PROBLEMS

1. $(x + 1)(x + 2)(x + 3)$

ANSWER $x^3 + 6x^2 + 11x + 6$

2. $(a + b)^2$

ANSWER $a^2 + 2ab + b^2$

3. $(a - b)^2$

ANSWER $a^2 - 2ab + b^2$

4. $(2x - 1)(x^2 + x - 3)$

ANSWER $2x^3 + x^2 - 7x + 3$

5. A rectangle $(x + 3)$ by $(x + 5)$: its area

ANSWER $x^2 + 8x + 15$

6. Its perimeter

ANSWER $4x + 16$

7. $(x + a)(x + b)$ in general

ANSWER $x^2 + (a + b)x + ab$

07 Homework**Homework — 5 tasks**

Count the products you expect before expanding, and again afterwards.

1. Expand $4x(2x - 3)$ and $-3a(a^2 + 2a - 5)$.
2. Expand $(x + 4)(x + 6)$ and $(x - 4)(x + 6)$.
3. Expand $(3a + 2)(2a - 5)$.
4. Expand $(x - 1)(x^2 + 2x + 3)$.
5. A rectangle has sides $x + 2$ and $x + 7$. Find its area and perimeter.